

AIC: balancing fit and complexity

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Last time: likelihood and deviance

- Return to the KLoSA data: **4,184 men aged 45 and older**.

```
klosa <- read_csv("klosa_men_grip_45plus.csv")
m_age <- glm(weak_grip ~ age,
  family = binomial, data = klosa)
summary(m_age)

#> Coefficients:
#>             Estimate Std. Error z value Pr(>|z|)
#> (Intercept) -8.824312    0.319252  -27.64  <2e-16 ***
#> age          0.116834    0.004755   24.57  <2e-16 ***
#> ---
#> Signif. codes:  0 '***' .001 '**' .01 '*' .05 '.' .1 ' ' 1
#>
#> (Dispersion parameter for binomial family taken to be 1)
#>
#>      Null deviance: 4259.0  on 4183  degrees of freedom
#> Residual deviance: 3455.1  on 4182  degrees of freedom
#> AIC: 3459.1
```

- The null model predicts the same probability, **0.206**, for every man.
- The residual deviance row is for the age model.
- The residual deviance is **803.9 points lower** than the null deviance.

```
anova(m_age, test = "LRT")

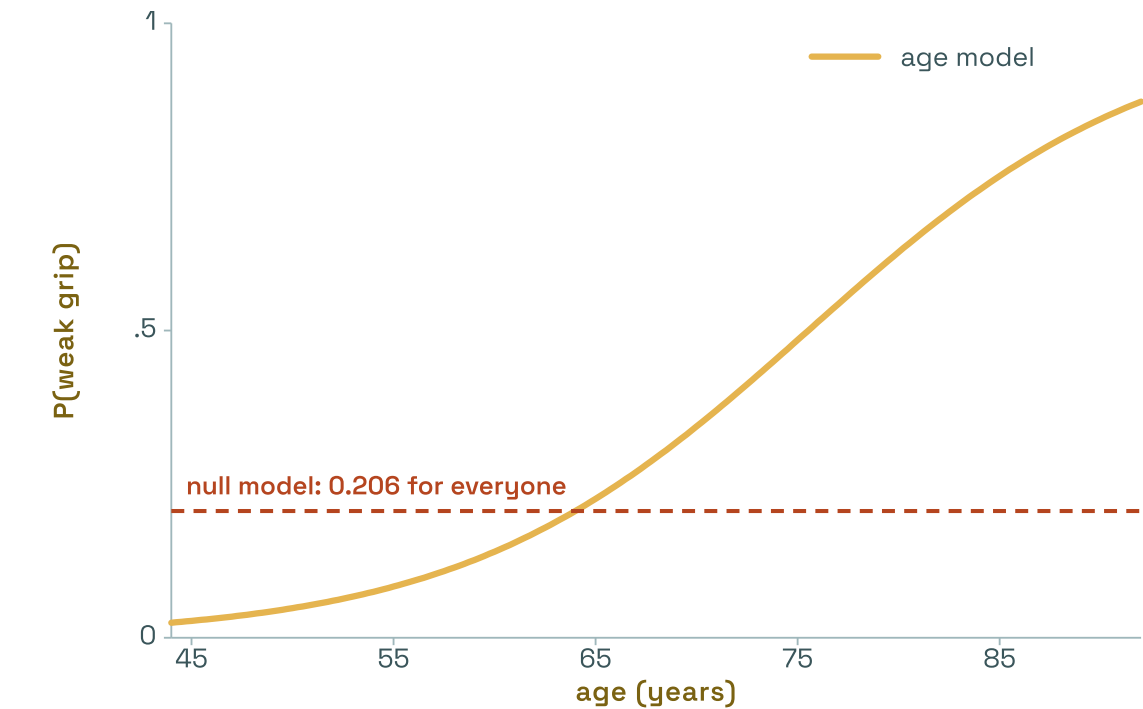
#>      Df Deviance Resid. Df Resid. Dev Pr(>Chi)
#> NULL          4183      4259.0
#> age     1      803.85      4182      3455.1 <2e-16 ***
```

- Information from an LRT often overlaps with whether coefficients are significant in `summary()`, so we leave it aside here. Unlike AIC, the standard LRT compares nested models only.
- Adding predictors almost always lowers residual deviance, even when they are useless. So we need to balance **better fit** against **added complexity**. That is the job of AIC.

PRACTICE Which model has residual deviance 3455.1?

- (a) `weak_grip ~ age` (b) `weak_grip ~ 1`

Answer: weak_grip ~ age



PRACTICE Why not choose a model using deviance alone?

- (a) more predictors usually lower it (b) larger is always better

Answer: more predictors usually lower it

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AIC rewards fit and penalizes complexity

- Akaike Information Criterion (AIC)** plays a role similar to **adjusted R²**: both reward fit while penalizing extra parameters.

$$AIC = \text{residual deviance} + 2k$$

Here, *k* counts coefficients including the intercept. Lower AIC is preferred.

AIC also works for linear models; we will cover that formula in another lesson.

```
set.seed(2026)
klosa <- klosa |> mutate(
  age_group = cut(age, c(45, 65, 75, Inf), include.lowest = TRUE),
  noise = rnorm(n()))
m0 <- glm(weak_grip ~ 1, family = binomial, data = klosa)
m_age <- glm(weak_grip ~ age, family = binomial, data = klosa)
m_noise <- glm(weak_grip ~ age + noise, family = binomial, data = klosa)
m_group <- glm(weak_grip ~ age_group, family = binomial, data = klosa)
```

```
deviance(m0); deviance(m_age); deviance(m_noise); deviance(m_group)
AIC(m0, m_age, m_noise, m_group)
```

```
#>      df deviance  AIC
#> m0      1  4259.0 4261.0
#> m_age    2  3455.1 3459.1
#> m_noise  3  3454.8 3460.8
#> m_group  3  3603.9 3609.9
```

	deviance		k / penalty	=	AIC
m0	4259.0	+	k=1 (+2)	=	4261.0
m_age	3455.1	+	k=2 (+4)	=	3459.1
m_noise	3454.8	+	k=3 (+6)	=	3460.8
m_group	3603.9	+	k=3 (+6)	=	3609.9

- An AIC of 3459.1 is not good or bad on its own.
- AIC becomes meaningful only when we compare candidate models.
- AIC is not a p-value and has no universal good/bad cutoff.

PRACTICE Which model does AIC prefer here?

- (a) `m_age`: 3459.1 (b) `m0`: 4261.0

Answer: `m_age`: 3459.1

PRACTICE AIC balances better fit against...

- (a) more parameters (b) a larger sample mean

Answer: more parameters

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Uses of AIC by model purpose

	Compare predictors Identify factors associated with outcome	Causal Estimate an exposure effect	Prediction Estimate future outcome risk
Compare functional forms Use the same variable, but as a linear term, quadratic term, or categories	✓ Home et al. 2018 Methods	✓ Orioli et al. 2018 Table 5	✓ Barrio et al. 2013 Table 1
Report fit across adjustment sets AIC compares relative fit	— usually fit one predictor model	✓ Chowsomchat et al. 2023 Table 3	— not usually described as adjustment sets
Use automated variable selection For example, forward or backward stepwise selection based on AIC	— use a specified predictor model	— confounders come from the DAG	✓ Whiteman et al. 2025 JAMA Dermatol.

Note: compare AIC only for the same outcome, observations, and likelihood family.

PRACTICE Can AIC compare a logistic model of weak grip (binary outcome) to a linear model of grip strength (continuous outcome)? (a) no (b) yes

Answer: no

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Learning objectives

YOU SHOULD NOW BE ABLE TO:

- ✓ Explain why **residual deviance alone** is not enough to choose a model.
- ✓ State the logistic regression **AIC formula**.
- ✓ Compare candidate models using AIC
- ✓ Name appropriate uses of AIC comparisons.